

# Constructing and Solving Equations — Paper C

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every equation on this paper is designed to come out exactly.
- The method is not printed on this paper. You are expected to know it.
- When a bracket is **subtracted**, the minus sign multiplies every term inside it. When a fraction wraps a whole side, multiplying by the denominator undoes it in one move.
- Several questions ask you to explain, prove or diagnose. A correct calculation is not by itself an explanation — say what the algebra shows.
- The mark for each question is shown in square brackets on the right.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

## SECTION 1 – BRACKETS, AND FRACTIONS

### 1. Solve:

(a)  $3(2x + 1) = 4x + 15$  [2]

---

---

---

(b)  $5(x - 2) = 3(x + 2)$  [2]

---

---

---

### 2. Solve:

(a)  $4(2x - 3) - 2(x - 5) = 22$  [2]

---

---

---

---

**(b)**  $3(x + 4) - 2(x - 1) = 20$

[2]

---

---

---

---

**3.** Solve:

**(a)**  $x \div 4 + 3 = 8$

[1]

---

---

**(b)**  $(x - 3) \div 5 = 4$

[1]

---

---

**(c)**  $2x \div 3 = 8$

[1]

---

---

**(d)**  $(2x + 1) \div 3 = 5$

[1]

---

---

SECTION 2 – NEGATIVES, AND COMPARISONS

**4.** Solve:

**(a)**  $7 - 2x = 1$

[2]

---

---

---

**(b)**  $10 - 3x = 4x - 4$

[2]

---

---

---

**5.** Two gyms charge differently.

**Gym A:** a 300 rupee joining fee, then 40 rupees a month.

**Gym B:** a 100 rupee joining fee, then 60 rupees a month.

**(a)** Write an expression for the total cost at each gym after  $m$  months. [1]

---

---

**(b)** Form and solve an equation to find after how many months the two totals are equal. [2]

---

---

---

**(c)** Which gym is cheaper over 15 months, and by how much? [1]

---

---

SECTION 3 – REASONING IT OUT

**6.** Three angles lie on a straight line. They measure  $(2x + 10)^\circ$ ,  $(3x - 5)^\circ$  and  $(x + 25)^\circ$ .

**(a)** Form an equation in  $x$ . State the angle fact you have used. [1]

---

---

**(b)** Solve it. [2]

---

---

---

**(c)** Work out the three angles, and check that they are consistent with your angle fact. [2]

---

---

---

7. Three consecutive whole numbers add up to 72.

(a) Using  $n$  for the smallest, form an equation.

[1]

---

---

(b) Solve it and write down the three numbers.

[2]

---

---

---

(c) Explain why three consecutive whole numbers can **never** add up to 70.

[2]

---

---

---

---

---

*End of paper. Check every answer has a unit or a form the question actually asked for.*